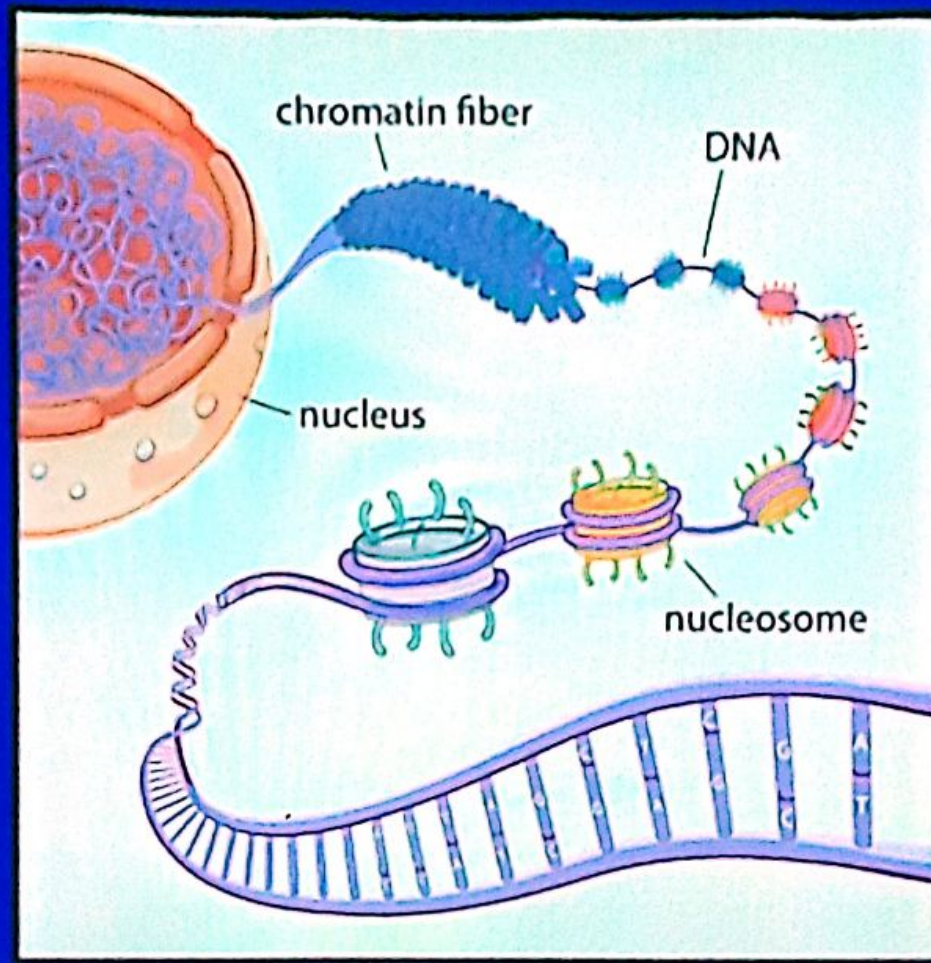


Chromatin

DNA is the main genetic constituent of the cells, carrying information in a coded form from cell to cell and from organism to organism

Within cells, DNA is complexed with Proteins in a structure called chromatin



- **Chromatin:** Contains **DNA**, **RNA**, basic proteins called **histones**, and **non-histone** (more acidic) proteins.
- **RNA** and **non-histone** protein is variable between different chromatin preparations
- **Histone** and **DNA** are always present in a fixed ratio of about 1:1

- **Non-histone proteins** are very heterogenous; they vary in different tissues and include **RNA and DNA polymerases**, other enzymes and regulatory proteins

There are only five types of histones, each one present in large amounts. This suggests that histones could have a structural role. Histones are small proteins that are basic because they have a high content (10 to 20%) of the basic amino acids arginine and lysine. Being basic, histones bind tightly to DNA, which is acid

- The four main histones, H2A, H2B, H3 and H4, are very similar in different species, being among the most conserved proteins known.
- Histone H1 is not conserved between species and has tissue-specific forms; is present only once per 200 base pairs of DNA.

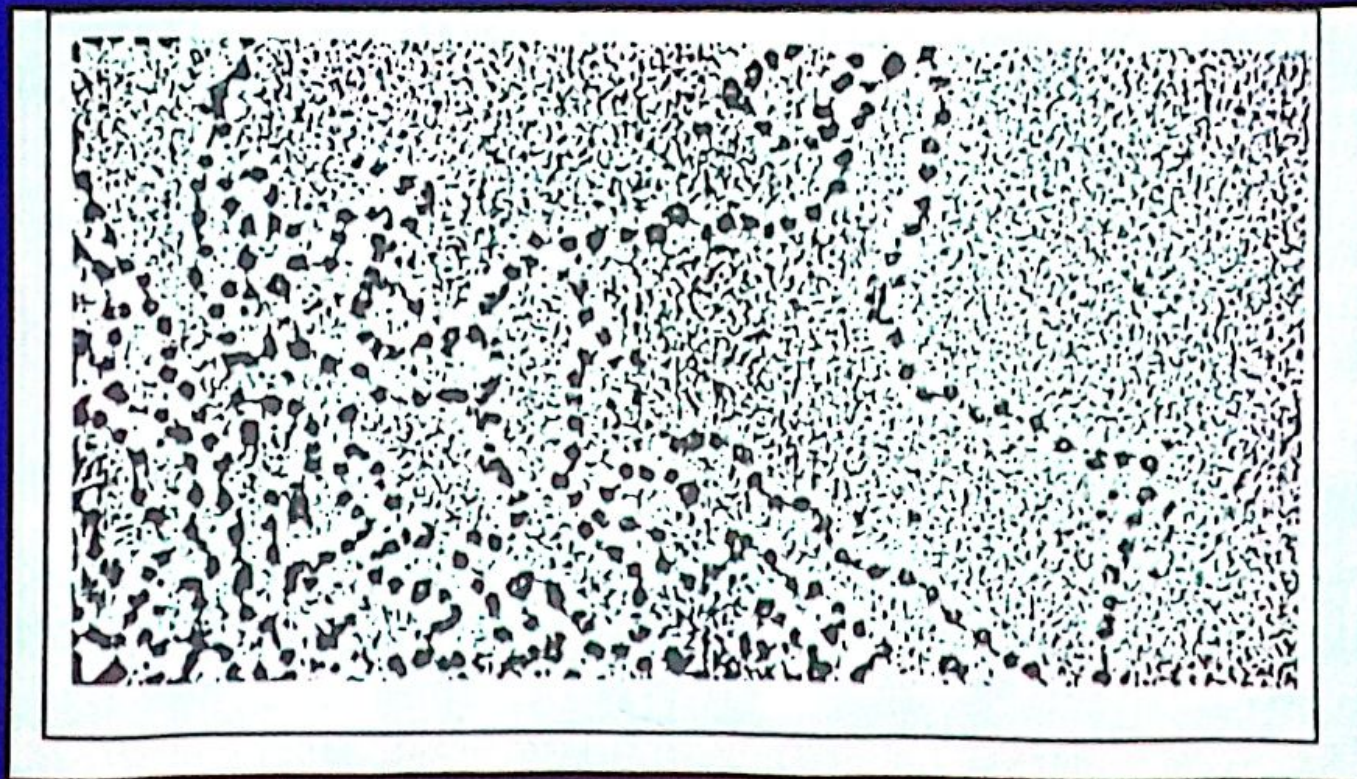
Composition of chromatin

Component	Liver	Pea embryo
DNA	31	31
RNA	5	17.5
Histone	36	33
Non-histone proteins	28	18

.Electron Microscopy (E.M.) of Chromatin

chromatin has a repeating structure of beads about 10 nm in diameter connected by a string of DNA.

The 10 nm fiber therefore represent the first level of organization of chromatin within cells.



Some properties of histones

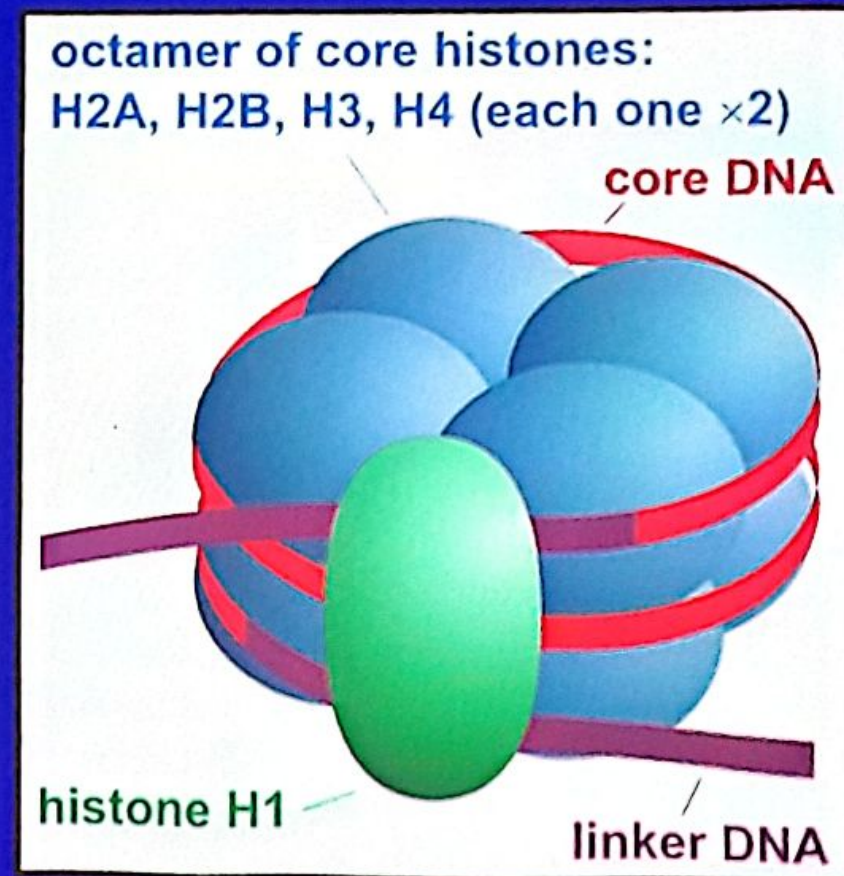
Histone	Molecular weight	acid Amino composition	Species variation	Number of molecules per 200 base pairs of DNA
H1	20,000	lysine-rich	Widly variable	1
H2A H2B	13,700 13,700	moderately-lysine rich	fairly well-conserved	2 2
H3 H4	10,700 11,200	arginine-rich	Highly conserved	2 2

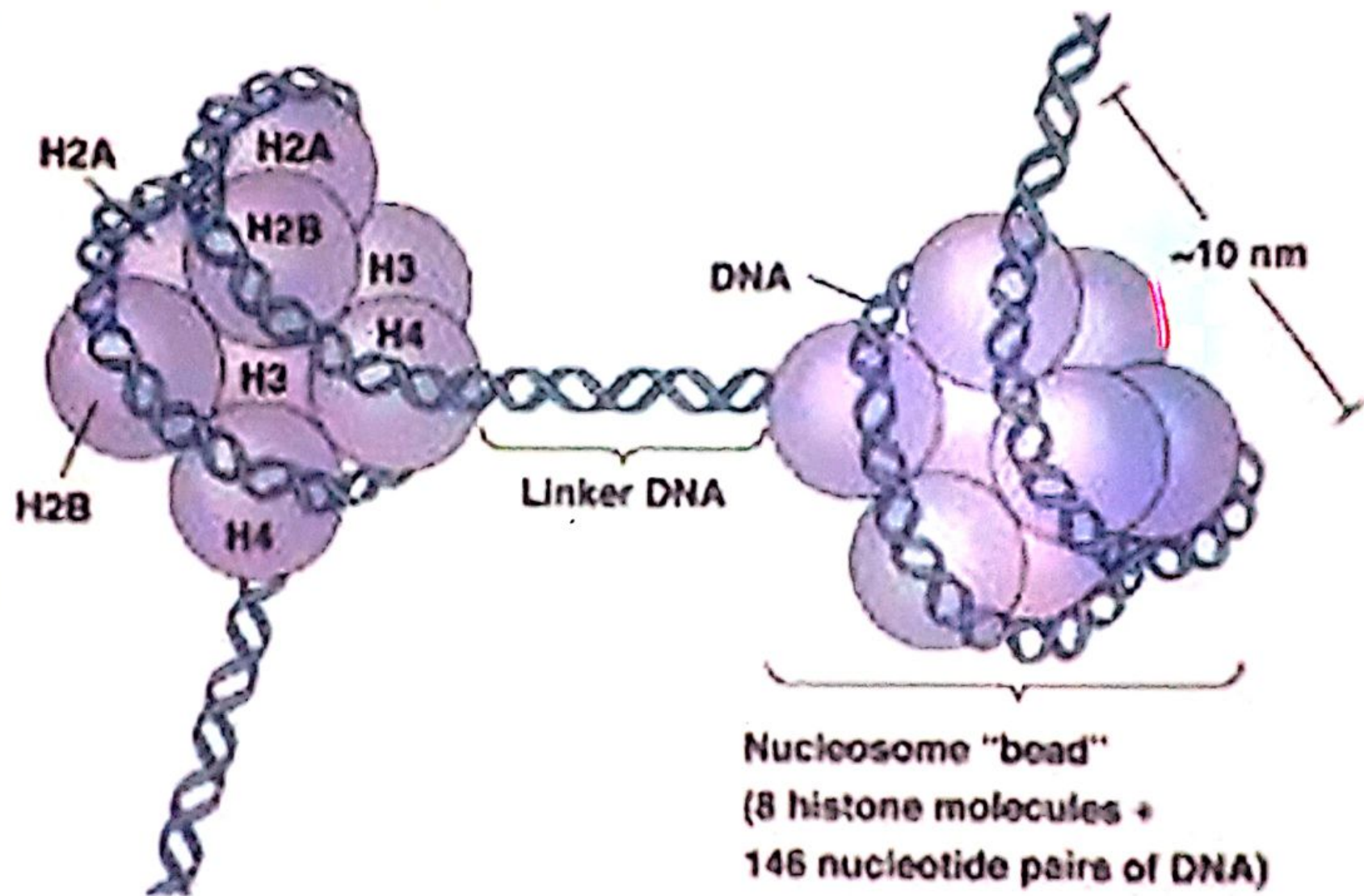
The Nucleosome

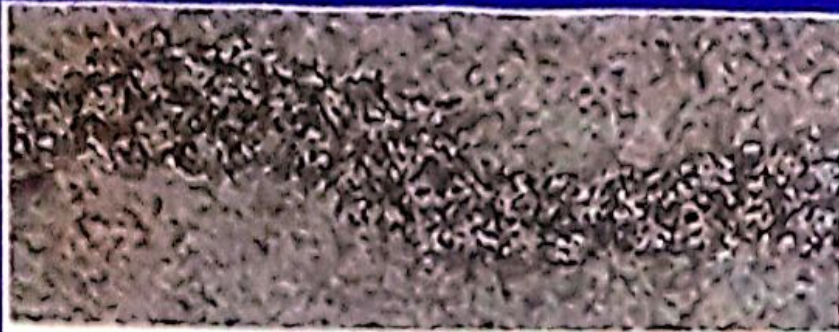
- The existence of a repeating unit of chromatin, called the *nucleosome*.

Four histones H2A, H2B, H3, and H4 are arranged in octamers containing two of each of them, every 200 base pairs of DNA.

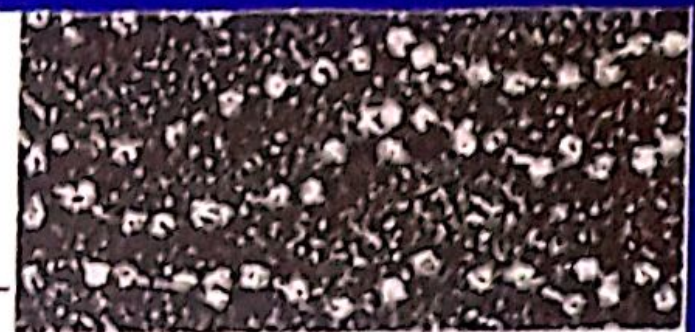
- Histone octamers are in close contact, and the DNA is coiled on the outside of the nucleosome



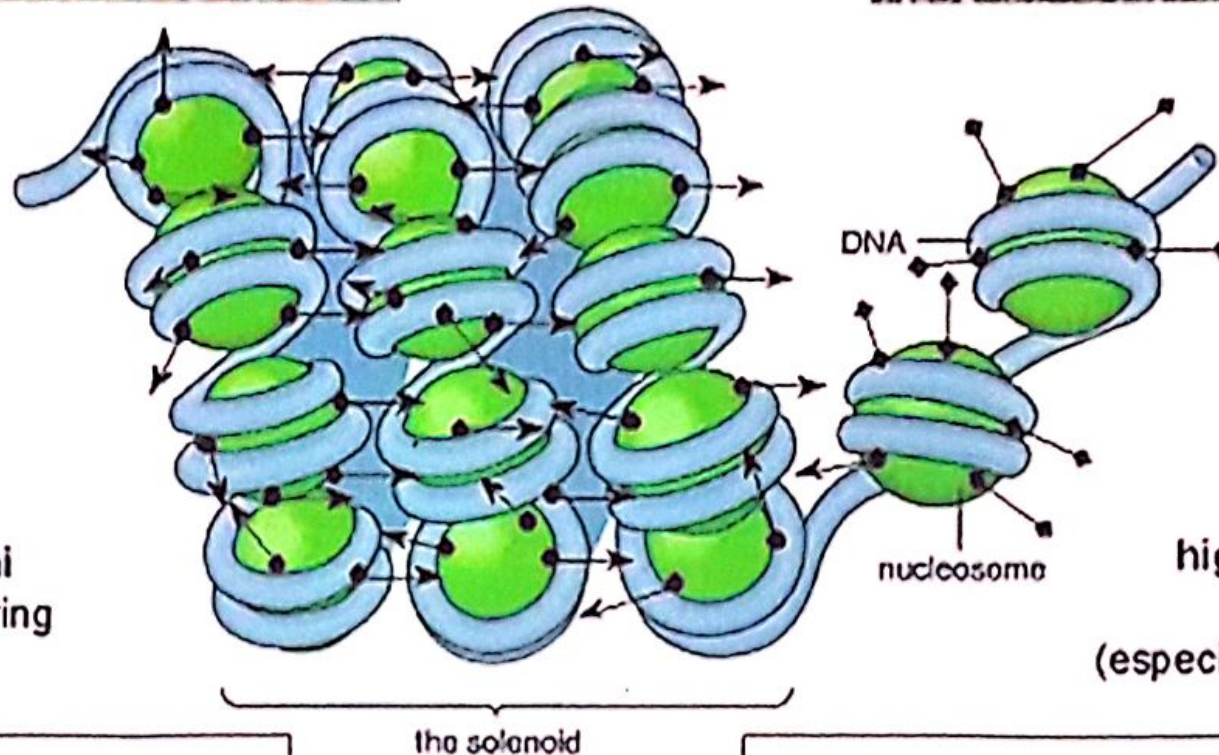




Chromatin fibers



30 nm
chromatin fiber



11 nm
(beads)



⊕ charged N termini
(bind DNA on neighboring
nucleosomes)



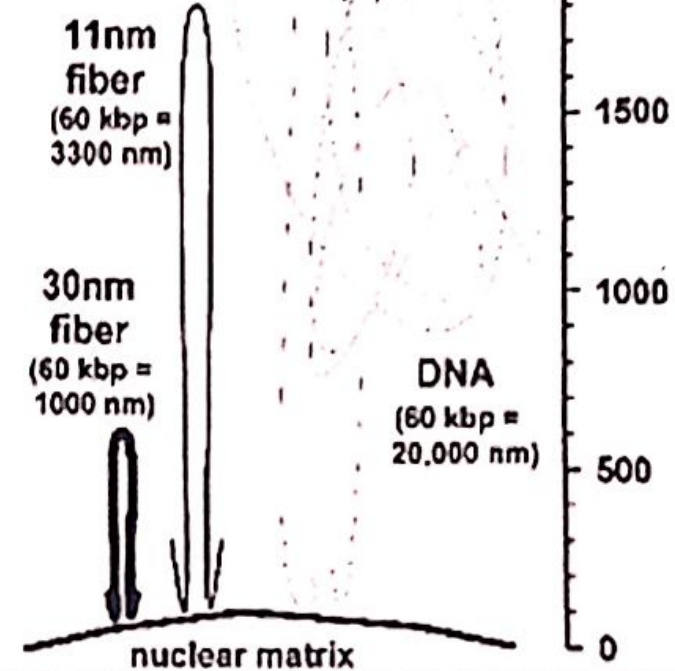
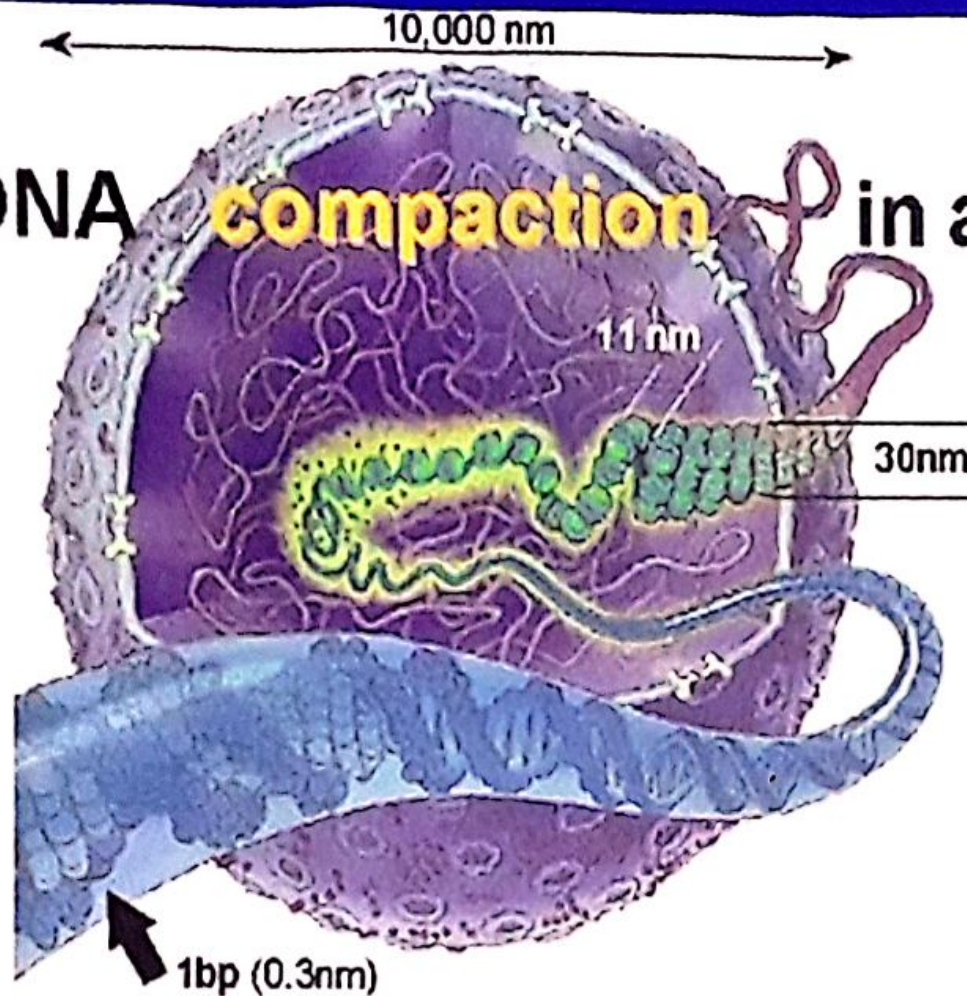
highly acetylated
core histones
(especially H3 and H4)

- HIGH level of histone H1
- NO gene transcription

- Reduced level of histone H1
- Gene transcription possible



DNA compaction in a human nucleus



			compact size	DNA	length	compaction
nucleus (human)	2 x 23 = 46 chromosomes	92 DNA molecules	10 μ m ball	12,000 Mbp	4 m DNA	400,000 x
mitotic chromosome	2 chromatids, 1 μ m thick	2 DNA molecules	10 μ m long X	2 x 130 Mbp	2 x 43 mm DNA	10,000 x
DNA domain	anchored DNA loop	1 replicon?	60 nm x 0.5 μ m	60 kbp	20 μ m DNA	35 x
chromatin fiber	approx. 6 nucleosomes per 'turn' of 11 nm		30 nm diameter	1200 bp	400 nm DNA	35 x
nucleosome	disk 1 1/4 turn of DNA (146 bp) + linker DNA		6 x 11 nm	200 bp	66 nm DNA	6 - 11 x
base pair			0.33 x 1.1 nm	1 bp	0.33 nm DNA	1 x



Compaction of DNA by histones

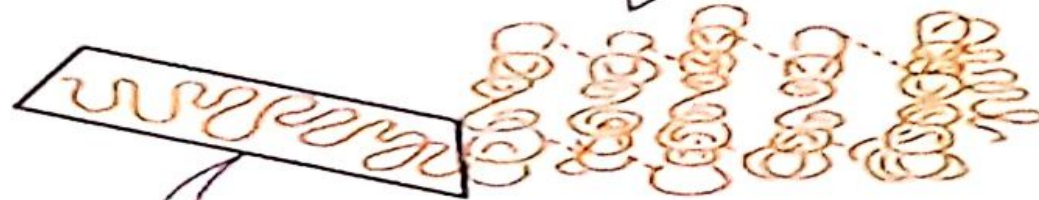
Compaction by chromosome scaffold / nuclear matrix

**Metaphase
chromosome**



1400 nm

**Condensed
scaffold-
associated
form**



700 nm

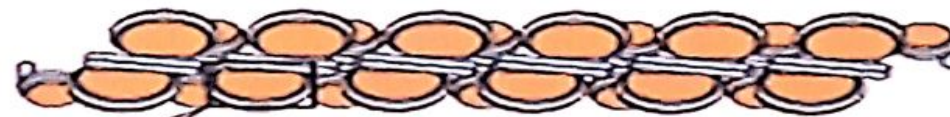
**Extended
scaffold-
associated
form**



Chromosome
scaffold

300 nm

**30-nm
chromatin
fiber of
packed
nucleosomes**



30 nm

**"Beads-
on-a-string"
form of
chromatin**

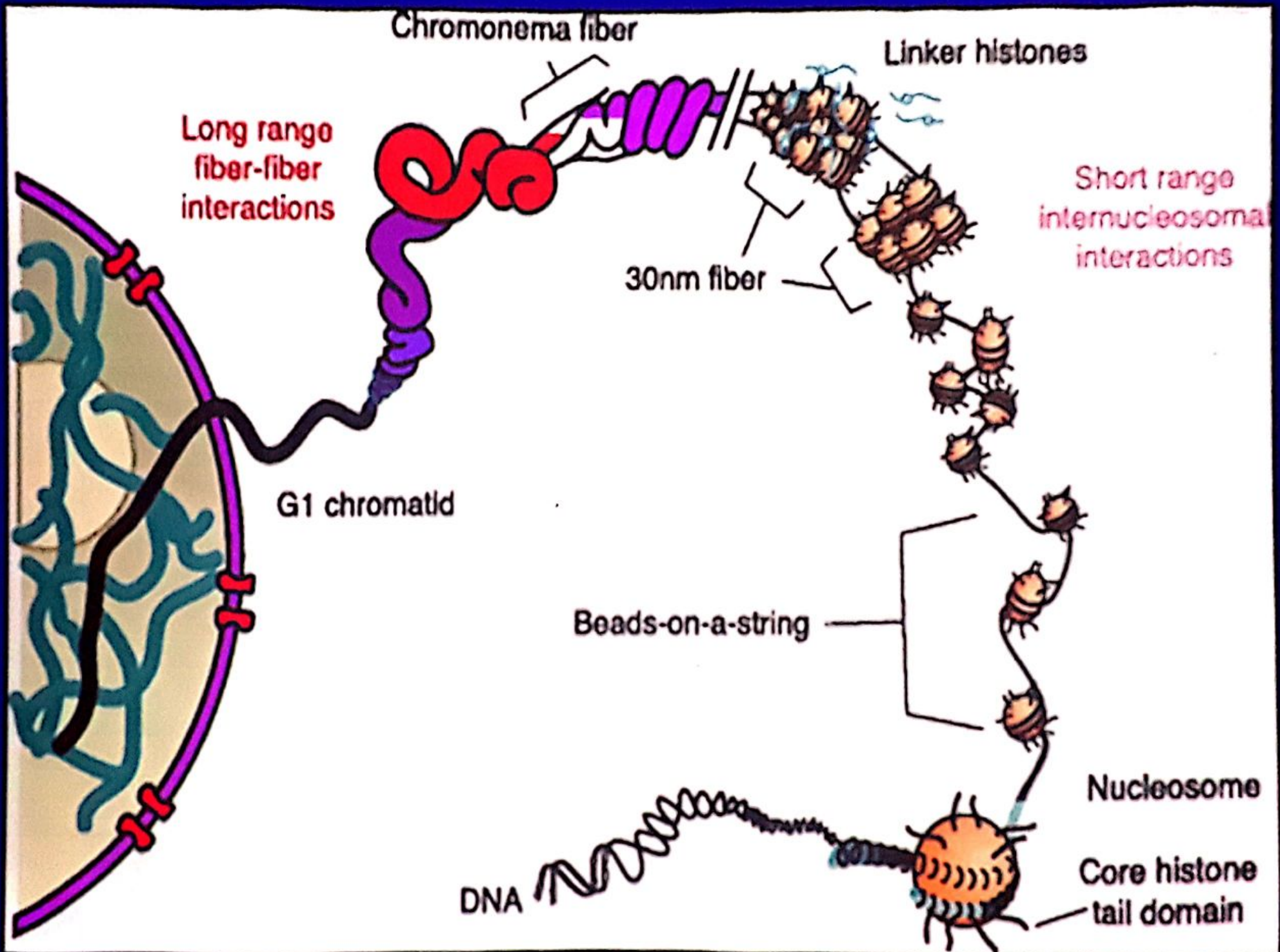


11 nm

**Short
region of
DNA double-
helix**



2 nm



chromatin is cleaved into discrete fragments by micrococcal nuclease.

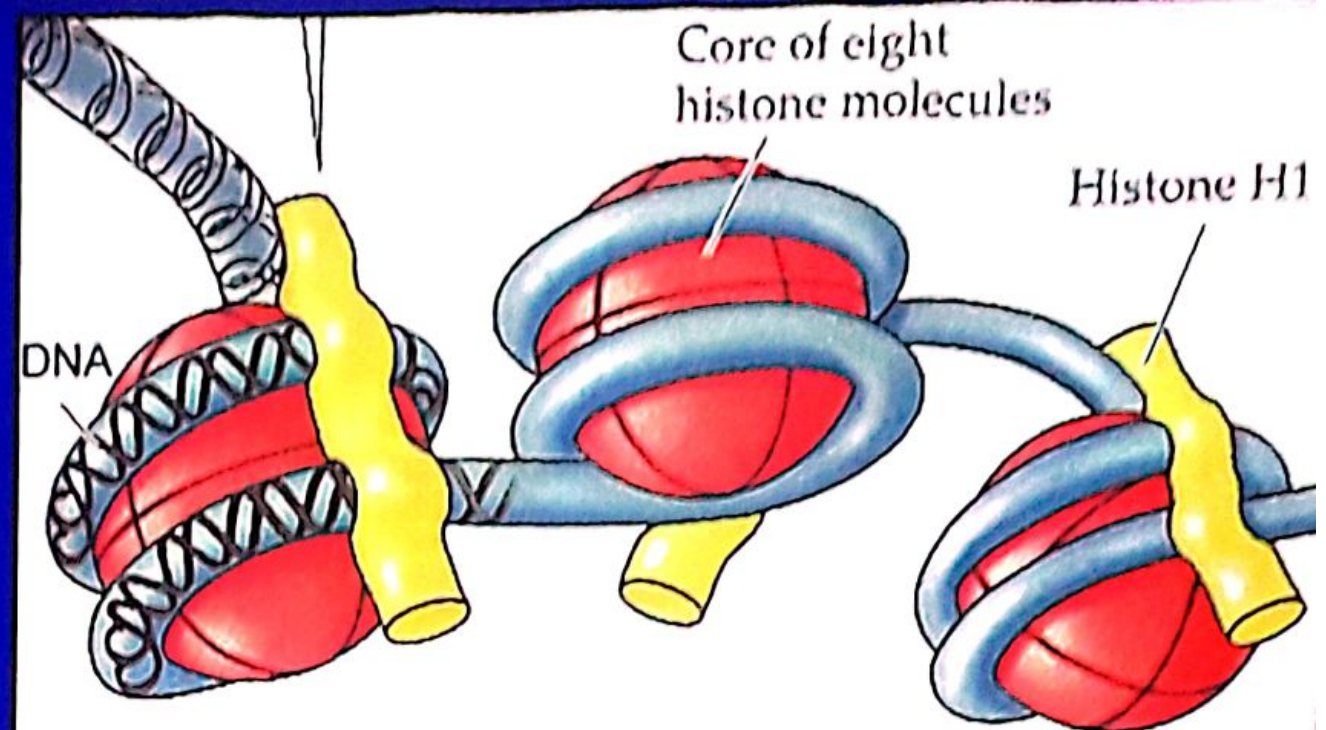
Neighbouring nucleosomes are connected by *linker* DNA, Brief digestion gives rise to *nucleosomes*, which contain a histone octamer, 200 base pairs of DNA and one molecule of histone H1.

Extensive digestion generates a nucleosome *core particle* that has lost the linker DNA and contains only 146 base pairs of DNA

When nucleosomes are in close apposition in the 10 nm filaments, the packing of DNA is about five- to seven-fold, i.e., five to seven times more compact than free DNA.

This is still 1000-fold lower than the packing ratio of DNA in metaphase chromosomes, which is achieved by further folding the 10 nm fiber .

Chromatin digestion



Nucleosome

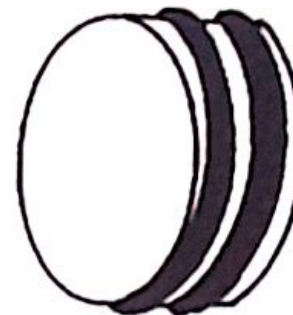
Chromatosome

Core Particle

165-240 bp +
octamer + H1

~166 bp +
octamer + H1

146 bp +
octamer



Bacterial Chromosomes Are Not Complexed with Chromatin

Prokaryotes do not have histones, and the DNA do not seem to be neutralized by basic proteins but rather by other substances, such as polyamines. On the other hand, all higher organisms have histones.

. In higher organisms, the chromatin undergoes condensation-decondensation cycles during cell proliferation, whereas bacterial DNA divides without changing its condensation state.

Chromatin Structure of Active Genes:

- Biochemical studies have shown that active genes are more accessible to nuclease digestion than are regions of the genome that are inactive.
- The chromatin of active genes is less condensed than that of inactive chromatin.
- in oviduct globin genes are inactive, the genes are not preferentially digested.
- Conversely, ovalbumin genes and their surrounding regions are DNase-sensitive in oviduct nuclei, but not in red blood cells.

The Mitotic Chromosomes

At the time of cell division, chromatin becomes condensed into the chromosomes..

Chromosomal Shape

Chromosomes can be studied in tissue sections, but they are best visualized in *squash* preparations.

Fragments of tissues are stained with basic dyes (e.g. orcein or Giemsa) and then squashed between slide and coverslip by gentle pressure.

The morphology of the chromosomes is best studied during metaphase and anaphase, which are the periods of maximal contraction.

Chromosomes are classified into four types according to their shape, which, in turn, is determined by the position of the centromere

Telocentric chromosomes: the centromere located on one end.

Acrocentric chromosomes: have a very small or even imperceptible short arm.

Submetacentric chromosomes: have arms of unequal length.

Metacentric chromosomes: have equal or almost equal arms.

N.B. During anaphase movements the chromosomes bend at the centromere.

so that Metacentric chromosomes are V-shaped and Acrocentric chromosomes are rod-shaped.

Metacentric

Short
arm p

Long
arm q



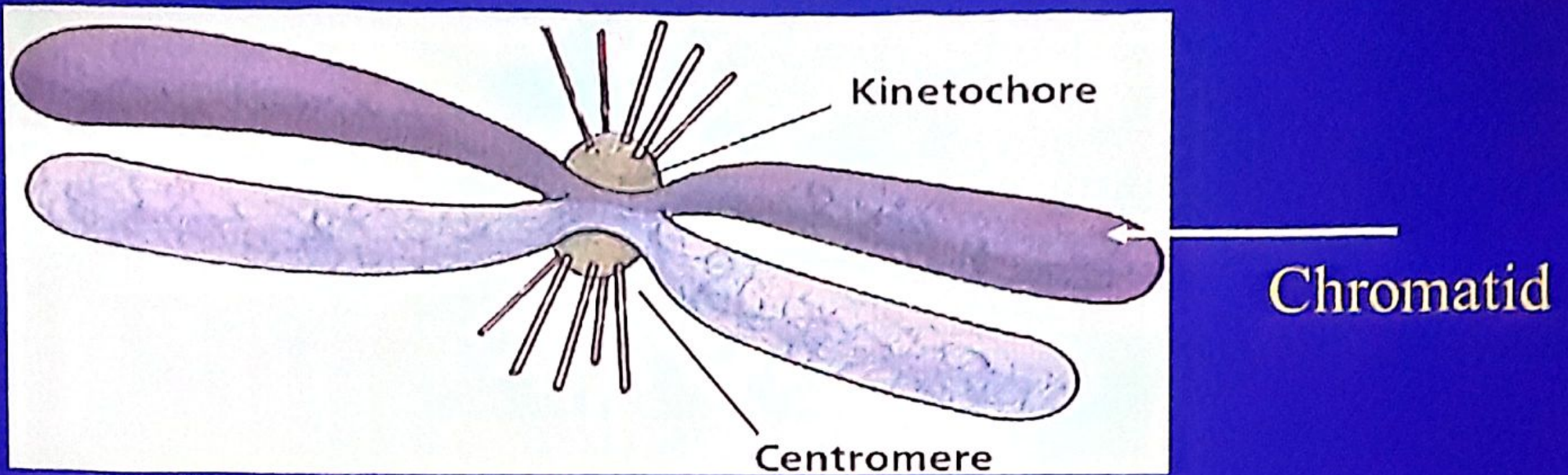
Submetacentric



Acrocentric



Chromosomal Nomenclature



Chromatid: Each chromosome consists of two symmetrical structures; the *chromatids* each one of which contains a single DNA molecule.

The chromatids are attached to each other only by the centromere and become separated at the start of anaphase.

Centromere (or Kinetochore).

This is the region of the chromosome that becomes attached to the mitotic spindle.

The centromere lies within the thinner segment of the chromosome, the *primary constriction*.

The region flanking the centromere frequently contain highly repetitive DNA (heterochromatin) and may stain more intensely with basic dyes.

Sometimes this disc, to which microtubules bind, is called the kinetochore.

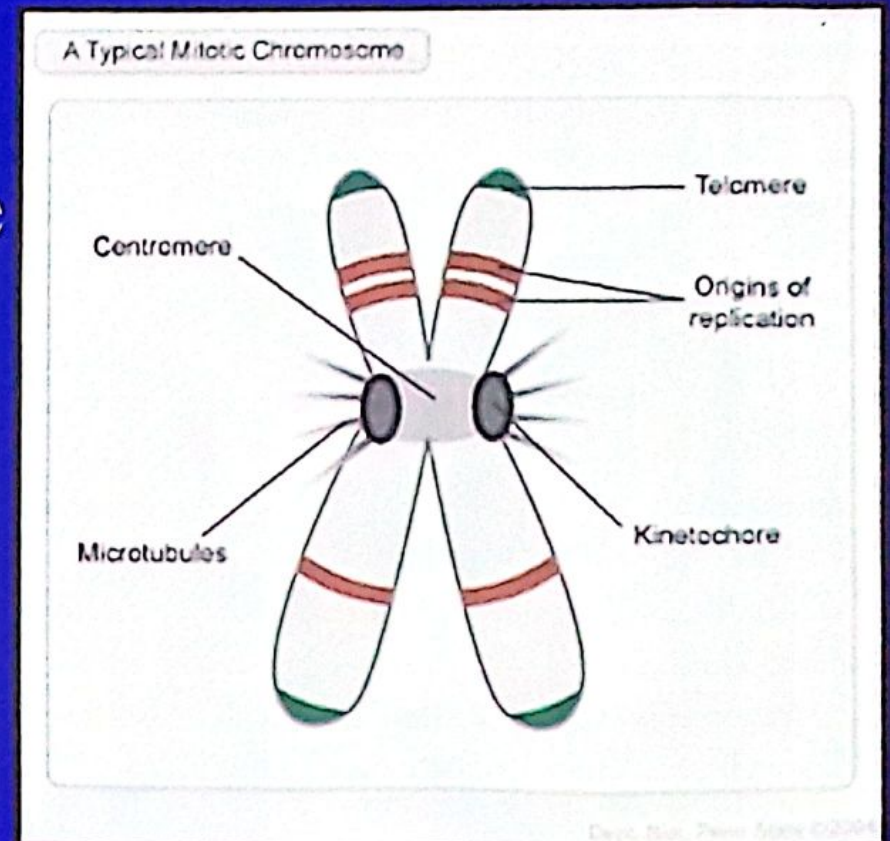
Between 4 and 40 microtubules become attached to the kinetochore and provide the force for chromosomal movement to one of the poles during the anaphase of mitosis and meiosis.

Telomere found at the end of linear chromosomes.

The function of telomeres is to provide stability to the chromosomes by rendering chromosome ends generally inert in interactions with other chromosomes ends.

Characters of telomeres:

1- Have an unusual DNA structure consists of short tandem repeats of DNA which is important for stability and integrity of the chromosome.



2- Replication of telomere requires a unique RNA-containing enzyme called telomerase.

In its absence, the DNA and the ends of the chromosome become shorter during each replication.

Because single-celled eukaryotes are immortalized cells, telomerase is critical for survival of such species.

In multicellular organisms, such as humans, telomerase is essential in germ line cells, but is inactive in most all types of somatic cells.

Chromosome shortening is considered part of the natural process of cell ageing, serving as an internal clock.

In human cancer cells, which have become immortalized, the transition to malignancy appears to require the activation of telomerase in order to overcome the normal senescence associated with chromosome shortening.

Secondary Constrictions.

Other morphological characteristics of chromosomes are the *secondary constrictions*.

Constant in their position and extent, these constrictions are useful in identifying particular chromosomes in a set.

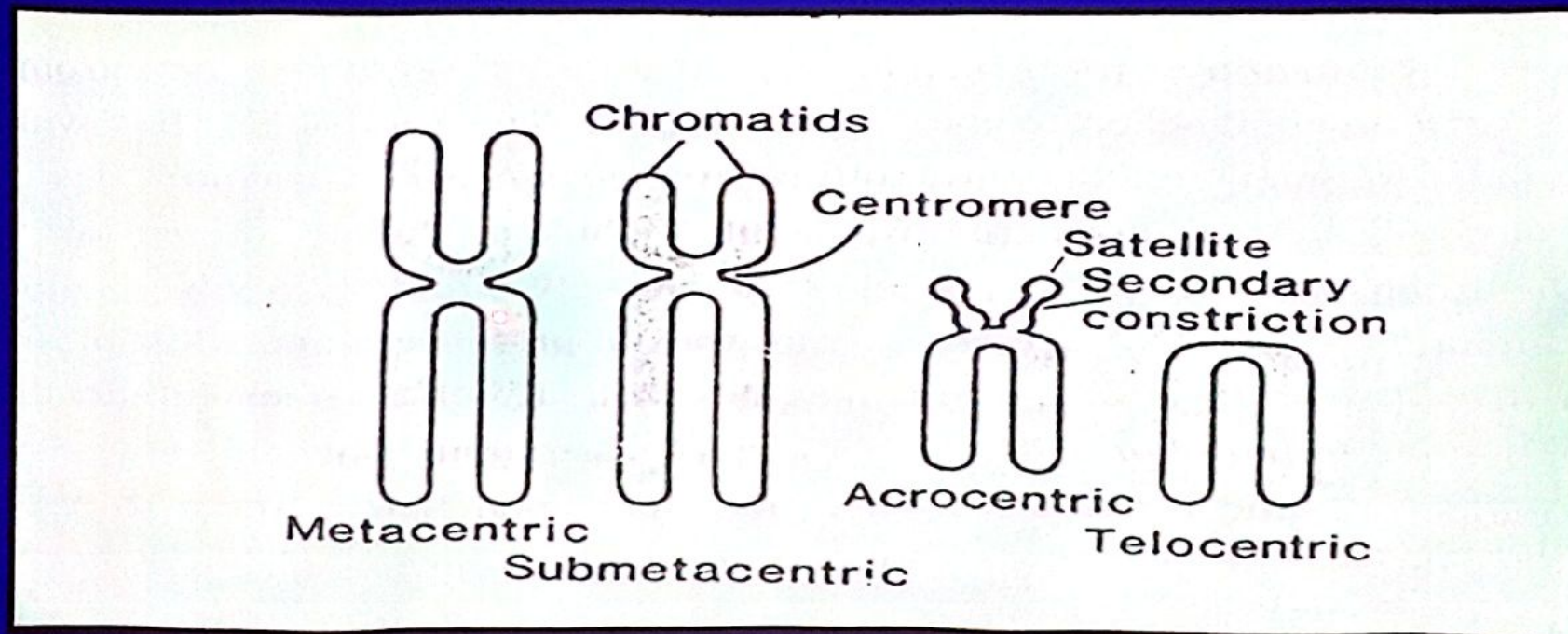
Nucleolar Organizers.

These areas are certain secondary constrictions that contain the genes coding for 18S and 28S ribosomal RNA and that induce the formation of nucleoli.

The secondary constriction may arise because the rRNA genes are transcribed very actively, interfering with chromosomal condensation.

Satellites. This is a rounded body separated from the rest of the chromosomes by a secondary constriction. The satellite and the constriction are constant in shape and size for each particular chromosome..

N.B. Chromosome satellites are a morphological entity and should not be confused with satellite DNAs, which are highly repeated DNA sequences.



Karyotyping

Karyotyping is a test to examine chromosomes in a sample of cells, which can help identify genetic problems as the cause of a disorder or disease. This test can:

- Count the number of chromosomes
- Look for structural changes in chromosomes.

During metaphase when the chromatids become shorter and thicker, the number, size and morphology of the chromosomes can be seen with the light microscope following appropriate treatment of the cells